Respiration

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core · Respiration · [1]

Answer: B

Explanation

Active transport, cell division and protein synthesis all use energy released by respiration. Diffusion and osmosis are passive processes that do not need energy from respiration, so options A, C and D each contain a process that does not belong.

2 · Part 1

Core · Aerobic respiration · [1]

Answer: D

Explanation

Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules to release energy. Option A is wrong because energy is released, not made, and it comes from the nutrient molecules, not from oxygen. Option B describes gas exchange and option C describes breathing (ventilation), which are not respiration.

3 · Part 1

Core · Aerobic respiration · [1]

Answer: A

Explanation

Aerobic respiration uses oxygen and produces carbon dioxide (glucose + oxygen → carbon dioxide + water), so oxygen decreases and carbon dioxide increases. Option B is the pattern for photosynthesis, which cannot happen in the dark. Options C and D would need both gases to change in the same direction, which does not match the equation.

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4 · Part 1 Supplement Aerobic respiration [1]

Answer: C

Explanation

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ is balanced: 6 carbon, 12 hydrogen and 18 oxygen atoms on each side. Option A is photosynthesis. Option B has 24 hydrogen atoms on the right but only 12 on the left, so it is not balanced. Option D is anaerobic respiration in yeast.

5 · Part 1 Core Anaerobic respiration [1]

Answer: A

Explanation

Anaerobic respiration is the chemical reactions in cells that break down nutrient molecules to release energy without using oxygen. Option B is wrong because anaerobic respiration does not use oxygen. Option C is wrong because many organisms, including humans and yeast, respire anaerobically when oxygen is short even though they live where oxygen is present. Option D is wrong because respiration happens in cells, not as a process of the lungs.

6 · Part 1 Core Anaerobic respiration [1]

Answer: D

Explanation

Anaerobic respiration in muscles uses no oxygen, releases much less energy per glucose molecule than aerobic respiration, and produces lactic acid only. Option A wrongly says more energy is released. Option B gives the products of aerobic respiration. Option C wrongly says oxygen is used.

7 · Part 1 Core Anaerobic respiration [1]

Answer: B

Explanation

Anaerobic respiration in yeast is glucose → alcohol + carbon dioxide. Options A and D give lactic acid, which is the product of anaerobic respiration in muscles, not yeast. Option C gives the products of aerobic respiration, which needs oxygen.

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8 · Part 1 Core Anaerobic respiration [1]

Answer: C

Explanation

Anaerobic respiration in muscles during vigorous exercise is glucose → lactic acid. Option A is anaerobic respiration in yeast. Option B is aerobic respiration, which needs oxygen. Option D is wrong because anaerobic respiration in muscles does not produce carbon dioxide.

9 · Part 1 Supplement Anaerobic respiration [1]

Answer: A

Worked solution

Ethanol : glucose = 2 : 1, so glucose broken down = 40 ÷ 2 = 20 molecules. Carbon dioxide : glucose = 2 : 1, so carbon dioxide produced = 2 × 20 = 40 molecules. Check: carbon dioxide and ethanol are produced in equal numbers, so 40 of each.

10 · Part 1 Supplement Anaerobic respiration [1]

Answer: B

Explanation

During vigorous exercise lactic acid builds up in the muscles and blood, causing an oxygen debt; extra oxygen is needed afterwards for the aerobic respiration of the lactic acid. Option A is wrong because carbon dioxide does not cause the oxygen debt. Option C is wrong because the debt is about removing lactic acid, not replacing glucose. Option D is wrong because human muscles do not make alcohol, and lactic acid is removed in the liver.

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11 · Part 1 Core Respiration [6]

Mark scheme

(a) **[1]**

- carbon dioxide (1)

REJECT oxygen; air

(b) **[3]**

- the number of bubbles increases as the temperature increases from 10 °C to 40 °C (1)
- the most bubbles are produced at 40 °C / the optimum is about 40 °C (1)
- the number of bubbles decreases above 40 °C / no bubbles are produced at 60 °C (1)
- maximum 2 marks if no data from the table are quoted (e.g. 72 bubbles at 40 °C, 0 bubbles at 60 °C)

(c) **[1]**

- 9.2 (bubbles per minute) (1)

(d) **[1]**

- the enzymes in the yeast are denatured / the yeast cells are killed (so they cannot respire) (1)

REJECT enzymes are killed

Worked solution

(a) Yeast respiration releases carbon dioxide gas, which escapes through the delivery tube as bubbles. (b) Read the table from top to bottom: the bubble count rises from 8 at 10 °C to a peak of 72 at 40 °C, then falls to 25 at 50 °C and to 0 at 60 °C. A full description gives the rise, the peak and the fall, with numbers from the table. (c) The bubbles were counted for 5 minutes, so the mean per minute is 46 ÷ 5 = 9.2. Check: 9.2 × 5 = 46. (d) Respiration is controlled by enzymes. Above the optimum temperature the enzymes change shape and are denatured, so the yeast stops respiring and no gas is produced. Enzymes are proteins, not living things, so they are denatured, not killed.

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12 · Part 1 Core Respiration [5]

Mark scheme

(a) **[2]**

- any two from: protein synthesis / cell division / active transport / growth / passage of nerve impulses / maintaining a constant body temperature (1 each, max 2)

REJECT diffusion; osmosis; breathing (this is muscle contraction)

(b) **[1]**

- active transport (1)

(c) **[2]**

- at 5 °C the mouse loses heat to its surroundings faster (1)
- so it needs to release more energy from respiration to keep its body temperature constant (1)

ALLOW respiration releases heat that keeps the body warm

REJECT respiration produces / makes energy

Worked solution

(a) Energy released by respiration is used for muscle contraction, protein synthesis, cell division, active transport, growth, the passage of nerve impulses and maintaining a constant body temperature. Any two apart from muscle contraction score. Diffusion and osmosis are passive, so they do not use energy from respiration. (b) Moving ions from a lower concentration in the soil to a higher concentration in the cell is against the concentration gradient, so it needs energy: this is active transport. (c) A mouse in a cold place loses heat to the air more quickly. To keep its body temperature constant it must replace that heat, so its cells respire faster and release more energy.

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13 · Part 1 Core Aerobic respiration [5]

Mark scheme

(a) **[2]**

- glucose + oxygen (1)
- → carbon dioxide + water (1)

ALLOW the correct balanced chemical equation

(b) **[2]**

- the germinating seeds in flask P respire (aerobically) and release carbon dioxide (which turns limewater milky) (1)
- the boiled seeds in flask Q are dead / do not respire, so no carbon dioxide is released (1)

ALLOW enzymes in the boiled seeds are denatured

(c) **[1]**

- respiration (in the germinating seeds) releases energy as heat (1)

REJECT respiration produces / makes energy

Worked solution

(a) Aerobic respiration: glucose + oxygen → carbon dioxide + water. (b) Limewater turns milky when carbon dioxide passes through it. All carbon dioxide was removed from the air before it entered the flasks, so any carbon dioxide reaching the limewater must come from the seeds. Germinating seeds are alive and respire, releasing carbon dioxide. Boiling kills the seeds, so they do not respire and release no carbon dioxide; flask Q is the control. (c) Some of the energy released by respiration is lost as heat. A vacuum flask keeps this heat in, so the temperature in P rises. The dead seeds do not respire, so the temperature in Q stays the same.

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14 · Part 1 Supplement Aerobic respiration [5]

Mark scheme

(a) **[2]**

- correct formulae: $C_6H_{12}O_6$ and O_2 on the left; CO_2 and H_2O on the right (1)
- correctly balanced: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (1)

ALLOW normal-size numbers for subscripts, e.g. C6H12O6

(b)(i) **[1]**

- 2.0×10^7 (1)

ALLOW 20 000 000

(b)(ii) **[1]**

- 1.2×10^8 (1)

ALLOW 120 000 000

(c) **[1]**

- 1 : 1 (1)

ALLOW 6 : 6

Worked solution

(a) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$. Check by counting atoms: 6 carbon, 12 hydrogen and 18 oxygen on each side. (b)(i) One glucose molecule uses 6 oxygen molecules, so glucose = $(1.2 \times 10^8) \div 6 = 2.0 \times 10^7$ molecules. Check: $2.0 \times 10^7 \times 6 = 1.2 \times 10^8$. (ii) Each glucose gives 6 water molecules, so water = $6 \times 2.0 \times 10^7 = 1.2 \times 10^8$. This equals the number of oxygen molecules, because both have the number 6 in the equation. (c) 6 molecules of carbon dioxide are produced for every 6 molecules of oxygen used, so the ratio is 6 : 6 = 1 : 1.

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15 · Part 1 Core Anaerobic respiration [5]

Mark scheme

(a) **[2]**

- max 2 from the following points:
- chemical reactions in cells (1)
- that break down nutrient molecules / glucose to release energy (1)
- without using oxygen (1)

(b) **[1]**

- anaerobic respiration releases much less energy (per glucose molecule) (1)

REJECT anaerobic respiration releases no energy

(c) **[2]**

- max 2 from the following points:
- less oxygen, so less aerobic respiration / more anaerobic respiration in the root cells (1)
- anaerobic respiration releases much less energy per glucose molecule (1)
- so less energy is available for active transport (1)

Worked solution

(a) Anaerobic respiration is the chemical reactions in cells that break down nutrient molecules to release energy without using oxygen. (b) It releases much less energy per glucose molecule than aerobic respiration. (c) Waterlogged soil has very little oxygen, so the root cells cannot respire aerobically as much and rely more on anaerobic respiration. This releases much less energy from each glucose molecule. Active transport needs energy from respiration, so with less energy available the cells take in mineral ions more slowly.

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16 · Part 1 Core Anaerobic respiration [6]

Mark scheme

(a) **[2]**

- glucose → (1)
- alcohol + carbon dioxide (1)

ALLOW ethanol for alcohol

(b) **[2]**

- (anaerobic) respiration of the yeast releases carbon dioxide (gas) (1)
- the gas is trapped as bubbles in the dough, so the dough expands / rises (1)

REJECT the yeast produces oxygen

(c) **[2]**

- increase = 340 − 200 = 140 (cm^3) (1)
- (140 ÷ 200) × 100 = 70 (%) (1)

ALLOW 70% with no working for 2 marks

Worked solution

(a) Anaerobic respiration in yeast: glucose → alcohol + carbon dioxide. (b) The yeast breaks down sugar by anaerobic respiration and releases carbon dioxide. The gas cannot escape easily from the stretchy dough, so it forms bubbles and the dough gets bigger. (c) Percentage increase = (increase ÷ starting value) × 100. Increase = 340 − 200 = 140 cm^3. Percentage increase = (140 ÷ 200) × 100 = 70%. Check: 200 + 70% of 200 = 200 + 140 = 340 cm^3. Always divide by the starting value, not the final value.

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17 · Part 1 Core Anaerobic respiration [5]

Mark scheme

(a) **[1]**

- glucose → lactic acid (1)

REJECT any equation with carbon dioxide or alcohol as a product

(b) **[2]**

- the muscles need a lot of energy very quickly / are contracting very fast (1)
- oxygen cannot be supplied (by the blood) fast enough for aerobic respiration to release all the energy needed (1)

(c) **[2]**

- more oxygen is taken in / delivered to the muscle cells (for aerobic respiration) (1)
- more glucose is delivered to the muscle cells / more carbon dioxide is removed from the muscle cells (1)

Worked solution

(a) Anaerobic respiration in muscles: glucose → lactic acid. (b) In a sprint the muscles contract very fast and need energy very quickly. The lungs and blood cannot supply oxygen to the muscle cells fast enough, so aerobic respiration cannot release all of the energy needed, and some cells respire anaerobically as well. (c) Faster, deeper breathing takes more oxygen into the blood. A faster heart rate pumps blood to the muscles more quickly, delivering more oxygen and glucose and carrying carbon dioxide away. This lets the muscle cells keep respiring aerobically.

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18 · Part 1 Supplement Anaerobic respiration [7]

Mark scheme

(a) **[2]**

- (during vigorous exercise) oxygen is not supplied to the muscles fast enough, so the muscles respire anaerobically (1)
- anaerobic respiration produces lactic acid, which passes from the muscles into the blood (1)

(b) **[3]**

- fast heart rate continues, to transport lactic acid in the blood from the muscles to the liver (1)
- deeper and faster breathing continues, to supply oxygen (for aerobic respiration of the lactic acid) (1)
- lactic acid is respired aerobically in the liver (1)

ALLOW lactic acid is broken down using oxygen in the liver

(c) **[2]**

- (11.0 − 2.0) ÷ 30 or 9.0 ÷ 30 (1)
- 0.30 mmol per dm^3 per minute (1)

ALLOW 0.3; mmol dm^{-3} min^{-1}

- the unit is needed for the second mark

Worked solution

(a) Running as fast as possible is vigorous exercise. The blood cannot bring oxygen to the muscles fast enough, so the muscles also respire anaerobically: glucose → lactic acid. The lactic acid builds up in the muscles and passes into the blood, so its concentration in the blood rises from 1.0 to 11.0 mmol per dm^3. (b) After exercise the heart rate stays high, carrying lactic acid in the blood from the muscles to the liver. Breathing stays deeper and faster, supplying extra oxygen. In the liver the lactic acid is respired aerobically, which removes it. (c) Decrease = 11.0 − 2.0 = 9.0 mmol per dm^3 over 40 − 10 = 30 minutes. Mean rate = 9.0 ÷ 30 = 0.30 mmol per dm^3 per minute. Check: 0.30 × 30 = 9.0, and 11.0 − 9.0 = 2.0.

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19 · Part 1 Supplement Anaerobic respiration [6]

Mark scheme

(a) **[2]**

- correct formulae: $C_6H_{12}O_6$ on the left; C_2H_5OH and CO_2 on the right (1)
- correctly balanced: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (1)

ALLOW normal-size numbers for subscripts

(b) **[2]**

- max 2 from the following points:
- the sugar / glucose is being used up (1)
- ethanol / alcohol builds up and is toxic to / kills the yeast (1)

ALLOW the number of living yeast cells has decreased

- carbon dioxide dissolves and lowers the pH, inhibiting the yeast's enzymes (1)

(c) **[2]**

- if oxygen were present, the yeast would respire aerobically (1)
- producing carbon dioxide and water instead of ethanol / so less ethanol would be made (1)

Worked solution

(a) $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. Check: 6 carbon, 12 hydrogen and 6 oxygen atoms on each side. (b) The temperature has not changed, so look at what changes in the tank. The yeast keeps using the sugar, so less substrate is left. Ethanol also builds up; at high concentrations it is toxic and kills the yeast. (c) Ethanol is only made by anaerobic respiration. With oxygen, the yeast would respire aerobically and the glucose would be broken down to carbon dioxide and water, so little or no ethanol would be produced.

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20 · Part 1 Supplement Anaerobic respiration [6]

Mark scheme

(a)(i) **[1]**

- 6 (1)

(a)(ii) **[1]**

- 2 (1)

(b) **[1]**

- 3 (times) (1)

ECF answer to (a)(i) ÷ answer to (a)(ii)

(c) **[1]**

- alcohol / ethanol (1)

ALLOW C_2H_5OH

(d) **[2]**

- anaerobic respiration releases much less energy per glucose molecule (than aerobic respiration) (1)
- so the yeast in Y must break down more glucose to release the same amount of energy (1)

Worked solution

(a) Aerobic: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ gives 6 carbon dioxide molecules per glucose. Anaerobic in yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ gives 2. (b) For the same number of glucose molecules, X produces 6 ÷ 2 = 3 times as many carbon dioxide molecules as Y. Check: for 10 glucose molecules, X makes 60 and Y makes 20, and 60 ÷ 20 = 3. (c) Only anaerobic respiration in yeast makes alcohol (ethanol). (d) Each glucose molecule broken down anaerobically releases much less energy than one broken down aerobically. To release the same amount of energy each hour, the yeast in Y must break down many more glucose molecules, so its glucose is used up faster.

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